NeoRepro 核心成果 | Bilingual Expert Brief

泄漏感知、患者级、可复现的 MHC-I 预测评测资源 | Leakage-aware, patient-level, reproducible benchmark resource

一句话结论

核心成果不是新模型或“普适赢家”，而是一套可复现证据链：固定版本、记录级溯源、训练重叠排除、共同支持、患者级统计与同支持随机基线。它能阻止外部验证和 Top-K 结果被误读。

BOTTOM LINE

The contribution is not a new model or universal winner, but an executable evidence chain: pinned versions, record provenance, overlap exclusion, common support, patient-level inference and support-matched random baselines. It prevents misleading external-validation and Top-K claims.

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TESLA 记录与 PRIME2 训练集精确重叠

All TESLA records exactly overlapped PRIME2 training data; retained only as a leakage-positive test fixture.

17,475

泄漏过滤后的 IMPROVE pMHC 记录

Leakage-filtered pMHC records: 465 positives, 70 patients and 3 cohorts on one common evaluation set.

2,315

过滤后的独立疫苗队列肽记录

Distinct post-vaccination ELISPOT endpoint: 311 positives from 352 patients after known-overlap exclusion.

证据示例 | EVIDENCE SNAPSHOT

support-aware references

IMPROVE 总体区分度 | Pooled AUROC

common support

BigMHC	0.546
PRIME	0.597

Zhao 患者级排序 | Patient NDCG@5

full support

BigMHC	0.658
Random	0.578

关键解释 | INTERPRETATION

Zhao gain over random: BigMHC +0.081; DeepHLApan +0.002; DeepImmuno -0.004 on 43.8% support.

实质价值 | WHY IT MATTERS

- **阻止错误外部验证**
  Prevents training-overlapped records from being presented as independent validation evidence.
- **让患者级比较可解释**
  Uses common support and support-matched random ranking so coverage and small candidate sets cannot inflate conclusions.
- **形成可扩展的公共资源**
  Packages five pinned predictors, canonical schemas, adapters, failures, overlap audits, metrics and a result manifest.

证据边界 | SCOPE

不是新预测器，不主张统计学成立的跨域排名反转，也不证明临床效益。

A benchmark/resource contribution - not natural presentation, killing or clinical benefit.

复现验证 | REPRODUCIBILITY

Independent clean clone: 32/32 tests; Ruff pass. Current metric audit: 34 checks; maximum error 2.2e-16.

主要来源 | KEY SOURCES

IMPROVE (2024) · Zhao vaccine cohort (2026) · PRIME2 (2023) · BigMHC (2023) · TESLA (2020)